The Supercomputer Entanglement

Ongoing use of PRC-Sanctioned Supercomputers by the Department of Defense and U.S. National Laboratories

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Executive Summary

Despite U.S. sanctions imposed in 2015, which aimed to restrict China's access to supercomputing capabilities, U.S. university researchers funded by the Department of Defense (DoD), along with researchers from Argonne (ANL), Oak Ridge (ORNL), and Los Alamos (LANL) National Laboratories, have continued to publish research findings using Chinese National Supercomputer Centers (NSCCs) that are listed on the Department of Commerce's Entity List. This ongoing collaboration highlights challenges in enforcing these restrictions effectively.[1]

Many co-researchers from the referenced scientific studies are affiliated with Chinese institutions that have clear ties to the People's Liberation Army (PLA), the Ministry of Industry and Information Technology (MIIT), and other state-run defense and strategic national programs. Notably, the National University of Defense Technology (NUDT), which reports directly to the Central Military Commission, has been the primary developer and manager of the TianHe-series supercomputers. These systems have been specifically implicated in activities that raise concerns about their use in defense and strategic contexts.[2] [3]

By merging U.S. expertise and research funding with PRC supercomputing resources—particularly the TianHe-series supercomputers in Tianjin and Guangzhou—these collaborations risk circumventing the 2015 restrictions, enabling unintended technology transfers, and undermining the sanctions' intended effect of restraining PLA weapons of mass destruction (WMD) program advancements.[4] An analysis of over 100 research papers from 2016 to 2024 reveals extensive collaboration between Chinese research institutions and U.S. universities funded by the DoD—including Los Alamos, Oak Ridge, and Argonne National Laboratories—with many papers explicitly referencing the TianHe-1, TianHe-1A, and TianHe-2 systems. This ongoing engagement exposes gaps in U.S. sanctions enforcement and, building on the Daily Caller News Foundation's original investigation, this report expands the findings through independent analysis.[5]

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1 Addition of Certain Persons to the Entity List; and Removal of Person From the Entity List Based on a Removal Request, A Rule by the Industry and Security Bureau on 02/18/2015, https://tinyurl.com/3jx7yxma, date accessed March 21, 2025
2 TianHe Series Supercomputer Systems, National University of Defense Technology, https://english.nudt.edu.cn/nav/Research/Achievements/6d10f012f67742ccb85ed8451dcaa710.htm, date accessed March 21, 2025
3 2022 Direct Recruitment Announcement of the College of Arts and Sciences of the National University of Defense Technology (国防科技大学文理学院2022年直招军官招聘公告)
https://web.archive.org/web/20220703174925/https://www.nudt.edu.cn/rczp/3624f026c40942088eb2ecb0ecdadf9a.htm, date accessed March 21, 2025
4 Tiffany Trader, Four Chinese Supercomputing Orgs Named on US Blacklist, HPC Wire, https://www.hpcwire.com/2015/04/08/chinese-supercomputing-orgs-placed-on-us-entity-list/, date accessed February 25, 2025
5 Philip Lenczycki, EXCLUSIVE: Pentagon, Energy Dept. Nuclear Research Projects Tapped Sanctioned Chinese Communist Party Supercomputers, https://dailycaller.com/2025/03/20/exclusive-pentagon-energy-dept-nuclear-research-projects-tapped-sanctioned-chinese-communist-party-supercomputers/, date accessed March 21, 2025

China's National Supercomputing Centers: The Backbone of PLA's Military-Technological Advancements

The NSCC's in China, including those in Tianjin and Guangzhou, serve as critical infrastructure for the PLA and the Chinese Communist Party (CCP) in advancing military technology, hypersonic weapons, and AI-driven defense capabilities. These centers house some of the world's most powerful supercomputers, notably Tianhe-2 at NSCC Guangzhou, developed by the National University of Defense Technology under the 12th Five-Year Plan 863 Program, which has been instrumental in supporting hypersonic missile research, aerospace simulations, satellite surveillance, cryptographic analysis, and nuclear weapons development. The NSCC Tianjin, home to Tianhe-1 and the next-generation Tianhe-3, plays a key role in PLA hypersonic propulsion simulations, strategic wargaming, and cyber warfare operations, while also advancing domestic high-performance computing (HPC) technologies like Feiteng processors and Kylin operating systems to reduce reliance on Western tech. The TianHe-1A has been constructed with an underground civil air defense data center engineered to withstand five conventional nuclear strikes. These supercomputers facilitate biomedical research, genetic analysis, air and space research, quantum cryptography, electronic warfare simulations, and counterspace operations, reinforcing China's civil-military fusion strategy. As part of the CCP's push for technological self-sufficiency, NSCCs continue to drive China's next-generation military advancements, ensuring computational superiority in hypersonic flight, space warfare, and AI-driven autonomous weapons systems, with broad implications for global security.[6] [7] [8] [9] [10] [11]

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6 National Supercomputer Center in Tianjin, Center Introduction, https://nscc-tj.cn/zxjj, date accessed March 21, 2025
7 National Supercomputer Center in Guangzhou, Basic Situation, http://www.nscc-gz.cn/Introduction/Generalprofile/Basicsituation.html, date accessed March 21, 2025
8 Yang, Y., Cheng, X., & Wang, W. (2016). 大规模流场模拟中壁面距离的并行计算方法研究 [Parallel computing method for wall distance calculation in large-scale flow field simulation]. J4, 38(06), 1086–1090. China PLA Unit 91550, Dalian, China; National University of Defense Technology (NUDT), Institute of Marine Science and Engineering, Changsha, China. http://joces.nudt.edu.cn/CN/abstract/abstract14882.shtml, date accessed March 21, 2025
9 徐佳庆,万文,吕启. Classification of Hyperspectral Remote Sensing Imagery Based on Second Order Moment Sparse Coding (基于二阶矩稀疏编码的高光谱遥感图像分类).计算机科学,2018,45(09):288-293. National University of Defense Technology; National Supercomputer Center in Guangzhou; Unit 31104 of PLA https://kns.cnki.net//kcms/detail/detail.aspx?filename=JSJA201809050&dbcode=CJFD&dbname=CJFDTEMP&v=-zevpYM5SKXfCAhUszTZuNOMC7KAeuzTJ4EpvVGlPQwynENEpeFG9ppatukyD-FI, date accessed March 21, 2025
10 Wang, R., Chen, Y., Peng, X., Cong, N., Fang, D., Liang, X. and Shang, J. (2022), "Topological design of the hybrid structure with high damping and strength efficiency for additive manufacturing", Rapid Prototyping Journal, Vol. 28 No. 10, pp. 1995-2003. https://doi.org/10.1108/RPJ-10-2021-0285, date accessed March 21, 2025
11 Luo, Y., Liu, J., Gong, C. et al. An efficient heterogeneous parallel password recovery system on MT-3000. J Supercomput 81, 38 (2025). https://doi.org/10.1007/s11227-024-06532-9, date accessed March 21, 2025

Key Observations

Widespread use of PRC-Sanctioned Supercomputers Among DoD Offices and U.S. National Labs

Between 2016 and 2024, 25 research papers funded by the DoD were identified, with 8 of these published from 2020 to 2024. Many of these papers explicitly referenced the TianHe-1, TianHe-1A, and TianHe-2 systems. This confirms the continued involvement with these platforms. Additionally, analysis of publications from national laboratories reveals 77 research papers during the same period, with 34 published after 2020. This significant usage of PRC supercomputing infrastructure, even after sanctions were imposed, underscores the complex challenges involved in preventing dual-use technology from enhancing China's strategic capabilities. These collaborations involves key DoD entities such as the Air Force Research Laboratory (AFRL), Air Force Material Command (AFMC), Naval Research Laboratory (NRL), Office of Naval Research (ONR), Army Research Laboratory (ARL), the Defense Advanced Research Projects Agency (DARPA), and the Defense Threat Reduction Agency (DTRA). Notably, 47 of these papers were published from 2020 to 2024, indicating that such interactions have persisted well after the implementation of 2015 sanctions. Many papers specifically referenced China's TianHe-1, TianHe-1A, and TianHe-2 supercomputing systems, highlighting continued use of these platforms. These partnerships, which span both fundamental and applied research, are particularly concerning in sensitive areas. Their ongoing nature, despite rigorous federal export control measures, underscores the significant challenges in fully enforcing these restrictions and highlights the enduring nature of international scientific collaborations.

Moreover, researchers funded by the DoD have also utilized other powerful Chinese computing resources, such as the Tianhe-2JK cluster at the Beijing Computational Science Research Center (CSRC) which is administered under the auspices of the China Academy of Engineering Physics (CAEP).[12] CAEP is China's main organization conducting the research, development, and testing of nuclear weapons and has been Entity Listed since 1997.[13] Others include the Sunway TaihuLight at the National Supercomputer Center in Wuxi, and the TianHe-1 at the National Supercomputer Center in Changsha.

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12 Beijing Computational Science Research Center Directors Message, https://www.csrc.ac.cn/en/about_csrc/directors_message/, date accessed March 21, 2025

13 Addition of Entities to the Entity List, Revision of Certain Entries on the Entity List
A Rule by the Industry and Security Bureau on 06/05/2020 , https://www.federalregister.gov/documents/2020/06/05/2020-10869/addition-of-entities-to-the-entity-list-revision-of-certain-entries-on-the-entity-list#page-34497, date accessed March 21, 2025

Universities Funded by the DoD Leveraging PRC Supercomputers
Several prestigious U.S. university researchers have been utilizing PRC supercomputers in conjunction with DoD funding. Researchers from institutions such as the Massachusetts Institute of Technology, University of California, Los Angeles, University of California, San Diego, Stanford University, University of Michigan–Ann Arbor, Georgia Institute of Technology, University of Texas at Austin, Princeton University, and University of Pennsylvania have acknowledged leveraging these powerful computing resources. These supercomputers were primarily used for computational processing and CPU time, critical elements supporting the universities' DoD-funded projects.

National Lab Divisions Leveraging PRC Supercomputers
Across Argonne, Oak Ridge, and Los Alamos National Laboratories, a diverse range of specialized divisions including X-ray science, neutron scattering, space science, and advanced computing have extensively utilized PRC supercomputers to propel their research forward. This collaboration spans various scientific fields, showcasing the role of high-performance computing in driving innovation and discovery in critical and strategic scientific domains. Below is a sampling of the divisions that have participated in these projects:

Argonne Divisions	Oakridge Divisions	Los Alamos Divisions
X-ray Science Division, Advanced Photon Source (APS)	Materials Science and Technology Division	Space Science and Application
Chemical Sciences and Engineering Division	Neutron Scattering Division	Theoretical Division
Materials Science Division	Quantum Condensed Matter Division	Physics and Chemistry of Materials
Center for Nanophase Materials Sciences (CNMS)	Center for Nanophase Materials Sciences	New Mexico Consortium
Physics Division	Computer Science and Mathematics Division	Center for Space Plasma Physics
	National Center for Computational Sciences	Center for Astrophysics
	Advanced Computing for Life Sciences and Engineering Group	Climate and Space Sciences and Engineering
	Chemical and Engineering Materials Division	

Dual-Use Research

Many of the studies center on fields with clear dual-use implications, including space weather modeling (critical for satellite defense and missile development), advanced materials science (e.g., quantum materials and metallic glasses), and computational simulations applicable to nuclear stockpile upkeep or hypersonic vehicle engineering. Even research traditionally viewed as benign—such as astrophysics—can offer insights that bolster defense or aerospace innovation, blurring the line between civilian and military applications. Additionally, collaborations utilizing DoD funding have expanded into specialized areas like neutron scattering and x-ray science, pivotal for understanding materials under extreme conditions—knowledge crucial for nuclear research and weapons development. These collaborative efforts not only enhance our understanding of complex scientific phenomena but also raise concerns about the potential for technological advancements to be utilized in enhancing military capabilities.

PLA and "Seven Sons" Affiliations

Numerous co-authors are affiliated with institutions directly tied to the PLA, such as the National University of Defense Technology and other universities collectively known as the "Seven Sons" of National Defense. Some of these organizations also operate under the Ministry of Industry and Information Technology, which oversees critical defense and aerospace initiatives. Additionally, the involvement of HPSTAR (High Pressure Science and Technology Advanced Research) and Beijing Computational Science Research Center, which are on the Entity List, further underscores the risk of inadvertently supporting Chinese military-industrial efforts.[14]

PRC National Funds and PRC Talent Programs Involved

U.S. University researchers funded by the DoD and researchers from Argonne, Oak Ridge, and Los Alamos engaged in projects that included key PRC funding initiatives such as the National Natural Science Foundation of China, National Key Research and Development Program of China, National Basic Research Program of China (973 program), Fundamental Research Funds for the Central Universities, Specialized Research Fund for State Key Laboratories, 111 Project, and the Space Application System of China Manned Space Program. Talent recruitment schemes like the Recruitment Program of Foreign Experts, Thousand Young Talents Program (Recruitment Program of Global Experts), Top 1000-Talents Award, National Youth Top-notch Talent Support Program, and the Young Elite Scientists Sponsorship Program by China Association for Science and Technology have also facilitated these research

[14] Addition of Entities to the Entity List, Revision of Certain Entries on the Entity List
A Rule by the Industry and Security Bureau on 06/05/2020 , https://www.federalregister.gov/documents/2020/06/05/2020-10869/addition-of-entities-to-the-entity-list-revision-of-certain-entries-on-the-entity-list#page-34497, date accessed March 21, 2025

collaborations that utilized these supercomputing resources. These programs align closely with the PRC's strategic interests in enhancing its technological sovereignty and military prowess.

Potential Loopholes

U.S. federal agencies and institutes—including the Department of Energy, Department of Defense, National Science Foundation, National Aeronautics and Space Administration, and National Institute of Standards and Technology—have unwittingly funded projects relying on PRC supercomputing, exposing lapses in oversight and enforcement. Existing grant and export-compliance frameworks often lack provisions requiring researchers to disclose HPC usage abroad, allowing sensitive work to proceed on prohibited systems. This lack of transparency raises pressing concerns over possible export-control violations and underscores the urgency of implementing more stringent review processes for international collaborations.

Unpacking the Dual-Use Implications

Example 1: In the study "Unraveling the dislocation–precipitate interactions in high-entropy alloys," collaboration between U.S. researchers from the University of Tennessee and their Chinese counterparts from the State Key Laboratory of Advanced Design and Manufacturing for Vehicle Body at Hunan University, along with the PRC Defense State Key Laboratory of Powder Metallurgy at Central South University, leveraged the computational capabilities of the TianHe-1 supercomputer located at the National Supercomputer Center in Changsha. This research delved into advanced materials science with clear dual-use implications, exploring the properties critical to military and aerospace applications, such as the development of stronger and more resilient materials for armor and spacecraft. The direct involvement of a laboratory under the PLA Defense Key Laboratory umbrella at Central South University highlights the dual-use nature of the findings, which are potentially applicable to defense-related material innovations. Furthermore, the University of Tennessee team's acknowledgment of support from the U.S. Army Research Office (contracts W911NF1310438 and W911NF1920049) underscores the strategic importance of this research.[15] [16] [17]

15 Li, J., Chen, H., Fang, Q., Jiang, C., Liu, Y., & Liaw, P. K. (2020). Unraveling the dislocation–precipitate interactions in high-entropy alloys. International Journal of Plasticity, 133, 102819. https://doi.org/10.1016/j.ijplas.2020.102819, date accessed March 21, 2025
16 Data Abyss PRC Defense Laboratories Tracker, https://www.dataabyss.ai/platform-resources/prc-defense-laboratories, date accessed March 21, 2025
17 U.S. Army CCDC Army Research Laboratory. (2020, April 23). Researchers create materials-by-design for future Army. U.S. Army. https://www.army.mil/article/234809/researchers_create_materials_by_design_for_future_army, date accessed March 21, 2025

Example 2: In the study "Effect of ion-specific water structures at metal surfaces on hydrogen production," researchers from Peking University and MIT used the TianHe-1A and TianHe-2 supercomputers to explore hydrogen production techniques, crucial nuclear energy applications. This work highlights the dual-use potential of hydrogen in nuclear fusion and fission processes, where it serves as fuel and coolant respectively. The research was funded by the National Key R&D Program of China, Strategic Priority Research Program of CAS, Key R&D Program of Guangdong Province, and the U.S. Army (contract W911NF1920065). The involvement of supercomputers known for their military affiliations in China raises significant security concerns about the inadvertent transfer of sensitive nuclear technologies.[18] [19]

Example 3: In the study "Polarity governs atomic interaction through two-dimensional materials," researchers from the Institute for Nuclear Energy and Technology at Sun Yat-sen University and U.S. researchers from MIT, MIT Lincoln Laboratory, University of Virginia, US Naval Research Laboratory, The Ohio State University and Georgia Institute of Technology utilized the National Supercomputer Center in Guangzhou to investigate atomic interactions in materials crucial for nuclear applications. This research was prominently supported by the China Scholarship Council and DoD programs, with funding from DARPA (contract 027049-00001), the Air Force Research Laboratory (contract FA94531820017), and the Office of Naval Research. These funding contributions alone highlight the study's strategic relevance to defense technologies.[20]

Example 4: In the study "Nucleon-Pair Shell Model within a Symmetry Broken Basis," Argonne researchers and their Chinese counterparts from Nankai University and Liaoning Normal University utilized the TianHe-1(A) supercomputer at the National Supercomputer Center in Tianjin to delve into advanced nuclear physics modeling. This research focuses on the nucleon-pair shell model, a critical method for understanding the arrangement and behavior of nucleons (protons and neutrons) in an atomic nucleus. Such investigations are pivotal for calculating nuclear energy levels, transitions, and reactions—key elements in both civilian nuclear energy and military nuclear technologies. By leveraging cutting-edge computational resources for this type of nuclear modeling, the collaboration poses risks due to the dual-use nature of the findings, which could

18 Tian, Y., Huang, B., Song, Y. et al. Effect of ion-specific water structures at metal surfaces on hydrogen production. Nat Commun 15, 7834 (2024). https://doi.org/10.1038/s41467-024-52131-w, date accessed March 21, 2025
19 IAEA (2023). Hydrogen production and nuclear energy applications. https://www.iaea.org/publications/8855/hydrogen-production-using-nuclear-energy, date accessed March 21, 2025
20 Kong, W., Li, H., Qiao, K. et al. Polarity governs atomic interaction through two-dimensional materials. Nature Mater 17, 999–1004 (2018). https://doi.org/10.1038/s41563-018-0176-4, date accessed March 21, 2025

inform nuclear weapons design and capabilities.[21] [22]

Example 5: In the study "Kohn Anomaly and Elastic Softening in Body-Centered Cubic Molybdenum at High Pressure," Argonne researchers and collaborators from the Center for High Pressure Science and Technology Advanced Research (HPSTAR) and the Institute of Atomic and Molecular Physics at Sichuan University— of which are on the U.S. Entity List—relied on the National Supercomputer Centre in Guangzhou (also Entity Listed) for high-pressure simulations. By partnering with these entity-listed organizations and deploying a sanctioned HPC resource, this project raises important concerns about whether advanced materials discoveries, crucial for aerospace or defense applications, may inadvertently benefit restricted PRC entities.[23] [24]

Example 6: In the study "Spin excitations in metallic kagome lattice FeSn and CoSn," researchers from Oak Ridge and several Chinese institutions such as the Beijing Key Laboratory of Opto-Electronic Functional Materials & Micro-Nano Devices of Renmin University of China and the Center for Advanced Quantum Studies and Department of Physics of Beijing Normal University investigate the magnetic properties of kagome metals—an area of intense interest for quantum materials and spintronic applications. The study leverages advanced neutron scattering at ORNL alongside high-performance simulations performed at the National Supercomputer Center in Guangzhou. This particular research highlights dual-use considerations, as the insights gained from these investigations could inform next-generation sensors, communication devices, and even quantum computing hardware with potential defense and aerospace implications.[25] [26]

Example 7: In the study "Improved Boundary Conditions for Coupled Geospace Models: An Application in Modeling Spacecraft Surface Charging Environment," Los Alamos researcher Vania K. Jordanova collaborates with Beihang University—one of the "Seven Sons" of China's defense sector—and the Key Laboratory of Space Environment Monitoring and Information Processing under the Ministry of Industry and Information

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21 He, B.; Zhang, S.; Zhang, Y.; Li, L.; Luo, Y.; Pan, F.; Draayer, J.P. Nucleon-Pair Shell Model within a Symmetry Broken Basis. Symmetry 2023, 15, 1653. https://doi.org/10.3390/sym15091653, date accessed March 21, 2025

22 Office of the Deputy Assistant Secretary of Defense for Nuclear Matters. (2020). Chapter 13: Basic nuclear physics and weapons effects. Nuclear Matters Handbook 2020 (Revised). U.S. Department of Defense. https://www.acq.osd.mil/ncbdp/nm/NMHB2020rev/chapters/chapter13.html, date accessed March 21, 2025

23 Yang, C., Zhang, Y., Salke, N. P., Bi, Y., Alatas, A., Said, A. H., Hong, J., & Lin, J.-F. (2022). Kohn anomaly and elastic softening in body-centered cubic molybdenum at high pressure. Physical Review B, 105, 094105. https://doi.org/10.1103/PhysRevB.105.094105, date accessed March 21, 2025

24 Research and Markets. (2023, April 27). Global molybdenum market report 2023: Usage in aerospace & defense bolsters sector growth. PR Newswire. https://www.prnewswire.com/news-releases/global-molybdenum-market-report-2023-usage-in-in-aerospace--defense-bolsters-sector-growth-301809760.html, date accessed March 21, 2025

25 Xie, Y., Chen, L., Chen, T., Wang, Q., Yin, Q., Stewart, J. R., Stone, M. B., Daemen, L. L., Feng, E., Cao, H., Lei, H., Yin, Z., MacDonald, A. H., & Dai, P. (2021). Spin excitations in metallic kagome lattice FeSn and CoSn. Communications Physics, 4, Article 240. https://doi.org/10.1038/s42005-021-00736-8 , date accessed March 21, 2025

26 Kang, Mingu, Ye, Linda, Fang, et al. Dirac fermions and flat bands in the ideal kagome metal FeSn. United States. https://doi.org/10.1038/s41563-019-0531-0, date accessed March 21, 2025

Technology (MIIT), which actively drives China's military-civil fusion initiatives. The study refines simulations for predicting spacecraft charging conditions—vital for both defense and commercial aerospace systems—and relies on the TianHe-2 supercomputer at the National Supercomputer Center in Guangzhou. As such, it exemplifies the dual-use dimension of space-plasma modeling, where advanced scientific research can simultaneously benefit military capabilities and civilian satellite operations.[27] [28]

Risks and Policy Implications

- **Technology Transfer**: Continued use of PRC supercomputers risks exposing advanced U.S. algorithms, modeling techniques, and other sensitive research outputs to Chinese entities linked to the PLA. Over time, these inadvertent transfers—whether through shared software repositories, publications, or direct technical exchanges—could bolster China's capacity in areas such as nuclear simulations, missile defense technologies, and hypersonic research. As a result, the work done at U.S. universities and national labs with taxpayers' dollars could be indirectly harnessed to accelerate China's military development and strategic edge.

- **Cybersecurity Risks:** The use of VPNs and SSH access by DOE and DoD-funded researchers to connect to PRC supercomputers presents serious cybersecurity risks, including potential interception by Chinese state-backed actors and exposure of U.S. research networks to malware due to weak security protocols at Chinese facilities. Additionally, these connections could facilitate unauthorized access, enabling cyber threats such as spear-phishing, social engineering attacks, and data exfiltration under the guise of legitimate research exchange. Lastly, reliance on Chinese supercomputing resources for processing U.S. research data raises legal and compliance risks, particularly concerning data sovereignty and exposure to foreign surveillance.

- **Evasion of Sanctions**: U.S. sanctions and export controls were implemented to curtail China's acquisition of cutting-edge supercomputing resources. However, the continued use of TianHe-series supercomputers and other PRC-based supercomputing systems raises the possibility that these measures are

27 Yu, Y., Su, S., Cao, J., Jordanova, V. K., & Denton, M. H. (2022). Improved boundary conditions for coupled geospace models: An application in modeling spacecraft surface charging environment. Space Weather, 20, e2022SW003178. https://doi.org/10.1029/2022SW003178 , date accessed March 21, 2025
28 Office of the NASA Chief Engineer. (2022, June 7). NASA technical handbook NASA-HDBK-4002B: Mitigating in-space charging effects—A guideline. NASA. https://standards.nasa.gov/sites/default/files/standards/NASA/B/0/2022-06-07-NASA-HDBK-4002B-Approved.pdf, date accessed March 21, 2025

insufficiently enforced or contain loopholes. By collaborating with Chinese supercomputing centers on the Department of Commerce's Entity List, U.S. researchers effectively circumvent the restrictions designed to limit the PLA's access to advanced computational power, undermining core U.S. foreign policy objectives.

- **Oversight Gaps**
  Evidence of DoD and DOE funding in joint research endeavors with Chinese supercomputing facilities underscores the complex oversight landscape facing federal agencies. Grant and contract administration processes and compliance reviews do not mandate transparent reporting on the use of sanctioned supercomputing resources. This lack of clarity and standardized enforcement opens avenues for sanctioned supercomputers to play an active role in federally funded projects, potentially undermining national security and eroding the purpose of export control mechanisms.

Recommendations for Congressional Consideration

- **Prohibit NDAA Funding for Research Using PRC High-Performance Computing:** Enact legislation to explicitly prohibit the use of National Defense Authorization Act (NDAA) funds for any research activities that utilize high-performance computing facilities from the PRC.
- **Strengthen Controls & Enforcement:** Close loopholes in federal research funding guidelines to prevent U.S. resources from supporting partnerships involving sanctioned PRC supercomputing facilities.
- **Mandate Comprehensive Disclosures:** Require U.S. researchers to disclose any foreign supercomputing resources used in federally funded projects, with particular attention to systems on the Entity List.
- **Enhance Compliance Reviews:** Instruct the DoD and DOE and other relevant agencies to intensify oversight of international collaborations, ensuring strict adherence to export controls and sanction requirements.
- **Audit Past and Current Projects:** Commission a thorough audit of research collaborations involving PRC supercomputers to identify any security breaches or sanction violations.

Conclusion

The Department of Defense and U.S. National Laboratories continued use of sanctioned Chinese supercomputers carries serious implications for American interests, particularly when collaborations involve entities affiliated with the People's Liberation Army and other core components of China's defense infrastructure. Allowing cutting-edge algorithms, data sets, and research methodologies to flow across institutional and national boundaries risks accelerating China's strategic capabilities. The lack of a uniform framework to monitor such collaborations underscores a broader policy gap, where federal grants, contracts, and export control measures may not effectively prevent the transfer of sensitive technology. Moving forward, Congress must consider legislative reforms, enhanced regulatory oversight, and comprehensive audits. Strengthening reporting requirements, mandating thorough review processes, and closing administrative loopholes are essential to prevent U.S.-funded research from inadvertently bolstering adversarial defense advances and to reaffirm America's leadership in high-performance computing while safeguarding its national security and foreign policy objectives.

Appendix A: Detailed Inventory of DoD-Funded Projects, Associated Contracts, and PRC Supercomputing Resources Leveraged

Research	Date	DoD Contracts Involved	PRC Supercomputers Used
Unlocking the Origin of High-Temperature Superconductivity in Molecular Hydrides at Moderate Pressures	11/1/2024	Department of Defense Research and Education Program for Historically Black Colleges and Universities and Minority-Serving Institutions Basic Research Funding (W911NF2310232).	TianHe-1A at National Supercomputer Center in Tianjin
Effect of ion-specific water structures at metal surfaces on hydrogen production	9/1/2024	Basic and Applied Scientific Research funded by the U.S. Army Research Laboratory; the U.S. Department of Defense (W911NF1920065).	TianHe-1A at National Supercomputer Center in Tianjin and TianHe-2 National Supercomputer Center in Guangzhou
Short-range interactions are irrelevant at the quasiperiodicity-driven Luttinger liquid to Anderson glass transition	01/01/2024	Air Force Office of Scientific Research (FA9550-20-1-0136).	Tianhe-2JK at Beijing Computational Science Research Center
Complex Materials with Stochastic Structural Patterns: Spiky Colloids with Enhanced Charge Storage Capacity	11/1/2023	Office of Naval Research Engineered Chiral Ceramics (N000141812876).	TianHe-1A at National Supercomputer Center in Tianjin
Unraveling the dislocation–precipitate interactions in high-entropy alloys	10/1/2020	Office of Naval Research Multidisciplinary University Research Initiative; Air Force Office of Scientific Research (N00014–17-1–2075; FA9550–16-1–0139).	TianHe-1 at National Supercomputer Center in Changsha
Hydration of NH4+ in Water: Bifurcated Hydrogen Bonding Structures and Fast Rotational Dynamics	9/1/2020	U.S. Army Research Office project (W911NF-13-1-0438; W911NF-19-2-0049).	TianHe-1A at National Supercomputer Center in Tianjin and TianHe-2 National Supercomputer Center in Guangzhou
Thiophene Cation Intercalation to Improve Band-Edge Integrity in Reduced-Dimensional Perovskites	06/01/2020	Air Force Office of Scientific Research (FA9550-16-1-0139).	TianHe-2 at National Supercomputer Center in Guangzhou

Molecular understanding of charge storage and charging dynamics in supercapacitors with MOF electrodes and ionic liquid electrolytes	02/01/2020	Office of Naval Research Multidisciplinary University Research Initiative (4-442521-JC-22891).	TianHe-2 at National Supercomputer Center in Guangzhou
Broadband transparent optical phase change materials for high-performance nonvolatile photonics	9/30/2019	U.S. Department of the Navy, Office of Naval Research (N00014-17-1-2524).	National Supercomputer Center in Guangzhou
Slater and Mott insulating states in the SU(6) Hubbard model	09/01/2019	Army Research Office (W911NF-17-1-0174).	TianHe-2 at National Supercomputer Center in Guangzhou
Interfacial oxygen-octahedral-tilting-driven electrically tunable topological Hall effect in ultrathin SrRuO3 films	07/01/2019	Air Force Contract; Defense Advanced Research Projects Agency through the Defense Sciences Office Program EXTREME Optics and Imaging; Young Faculty Award Program (FA8721-05-C-0002 or FA8702-15-D-0001; HR00111720029; D18AP00070).	TianHe-1 at National Supercomputer Center in Changsha
Identity of attached eddies in turbulent channel flows with bidimensional empirical mode decomposition	05/01/2019	Air Force Office of Scientific Research (FA9550-14-1-0168).	TianHe-2 at National Supercomputer Center in Guangzhou
Formation and Healing of Defects in Atomically Thin GaSe and InSe	04/01/2019	Army Research Office Young Investigator Program Award (W911NF1810198).	TianHe-2 at National Supercomputer Center in Guangzhou
Probing the phase transformation and dislocation evolution in dual-phase high-entropy alloys	3/1/2019	Office of Naval Research (N00014-16-S-BA10).	TianHe-1 at National Supercomputer Center in Changsha
Polarity governs atomic interaction through two-dimensional materials	10/8/2018	Defense Threat Reduction Agency Grant (HDTRA1-12-1-0013).	National Supercomputer Center in Guangzhou

Type-II Multiferroic Hf2VC2F2 MXene Monolayer with High Transition Temperature	7/11/2018	Defense Advanced Research Projects Agency (027049-00001).	National Supercomputer Center in Guangzhou
Mott transition in the π-flux SU(4) Hubbard model on a square lattice	05/01/2018	Naval Research Laboratory supported by the Office of Naval Research.	TianHe-2 at National Supercomputer Center in Guangzhou
Observation of Terahertz Radiation via the Two-Color Laser Scheme with Uncommon Frequency Ratios	12/8/2017	Army Research Office (W911NF-16-1-0255).	TianHe-2 at National Supercomputer Center in Guangzhou
Raman spectroscopy and x-ray diffraction of sp3CaCO3 at lower mantle pressures	9/1/2017	Air Force Office of Scientific Research (FA9550-14-1-0168).	TianHe-2 at National Supercomputer Center in Guangzhou
Eigenstate thermalization in the two-dimensional transverse field Ising model. II. Off-diagonal matrix elements of observables	07/01/2017	Army Research Office (US ARMY W911NF-17-1-0428).	Tianhe-2JK at Beijing Computational Science Research Center
A stable compound of helium and sodium at high pressure	02/01/2017	Army Research Office (56122-CH-H).	TianHe-2 at National Supercomputer Center in Guangzhou
Finite-temperature valence-bond-solid transitions and thermodynamic properties of interacting SU(2N) Dirac fermions	02/01/2017	U.S. Office of Naval Research (N00014-14-1-0540).	TianHe-2 at National Supercomputer Center in Guangzhou
The phase behavior study of human antibody solution using multi-scale modeling	11/01/2016	Defense Advanced Research Projects Agency (W31P4Q1210008).	TianHe-1A at National Supercomputer Center in Tianjin
Regulation of actin catch-slip bonds with a RhoA-formin module	10/12/2016	Defense Threat Reduction Agency (HDTRA1-14-1-0017).	National Supercomputing Center in Tianjin

Non-localized deformation in CuZr multi-layer amorphous films under tension	9/1/2016	Army Research Office (W911NF-16-1-0257).	TianHe-1A at National Supercomputer Center in Tianjin

Appendix B: Detailed Inventory of DOE-Funded Projects, Associated Personnel, and PRC Supercomputing Resources Leveraged

Research	Date	DOE Personnel Involved	PRC Supercomputers Used
Global Distribution of EMIC Waves and Its Association to Subauroral Proton Precipitation During the 27 May 2017 Storm: Modeling and Multipoint Observations	5/29/2024	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Stereoselective recognition of morphine enantiomers by μ-opioid receptor	1/22/2024	Ngo, Van A (Advanced Computing for Life Sciences and Engineering Group, Science Engagement Section, National Center for Computational Sciences, Oak Ridge National Lab)	TianHe Supercomputer at National Supercomputer Center in Tianjin
Signature of spin-phonon coupling driven charge density wave in a kagome magnet	10/4/2023	Miao, H. (Materials Science and Technology Division, Oak Ridge National Laboratory), Li, H. X. (Materials Science and Technology Division, Oak Ridge National Laboratory; Advanced Materials Thrust, The Hong Kong University of Science and Technology (Guangzhou)), Fabbris, G. (Advanced Photon Source, Argonne National Laboratory), Said, A. H. (Advanced Photon Source, Argonne National Laboratory), Tartaglia, R. (Advanced Photon Source, Argonne National Laboratory; "Gleb Wataghin" Institute of Physics, University of Campinas), Okamoto, S. (Materials Science and Technology Division, Oak Ridge National Laboratory), Lee, H. N. (Materials Science and Technology Division, Oak Ridge National Laboratory)	TianHe-1A at National Supercomputer Center in Tianjin

Nucleon-Pair Shell Model within a Symmetry Broken Basis	8/26/2023	He, Bingcheng (Physics Division, Argonne National Laboratory)	TianHe-1A at National Supercomputer Center in Tianjin
Kanatzidisite: A Natural Compound with Distinctive van der Waals Heterolayered Architecture	8/8/2023	Zhou, Xiuquan (Materials Science Division, Argonne National Laboratory)	National SuperComputer Center in Tianjin
Numerical Modeling of Energetic Electron Acceleration, Transport, and Emission in Solar Flares: Connecting Loop-top and Footpoint Hard X-Ray Sources	12/15/2022	Guo, Fan (Los Alamos National Laboratory; New Mexico Consortium)	TianHe-2 at National Supercomputer Center in Guangzhou
Improved Boundary Conditions for Coupled Geospace Models: An Application in Modeling Spacecraft Surface Charging Environment	9/1/2022	Jordanova, Vania K. (Space Science and Applications, Los Alamos National Laboratory), Denton, Michael H. (New Mexico Consortium)	TianHe-2 at National Supercomputer Center in Guangzhou
A Model of Double Coronal Hard X-Ray Sources in Solar Flares	7/6/2022	Guo, Fan (Los Alamos National Laboratory; New Mexico Consortium)	TianHe-2 at National Supercomputer Center in Guangzhou
Atomically dispersed Pt and Fe sites and Pt–Fe nanoparticles for durable proton exchange membrane fuel cells	6/1/2022	Xu, Gui-Liang (Chemical Sciences and Engineering Division, Argonne National Laboratory), Hwang, Inhui (X-ray Science Division, Argonne National Laboratory), Sun, Cheng-Jun (X-ray Science Division, Argonne National Laboratory), Amine, Khalil (Chemical Sciences and Engineering Division, Argonne National Laboratory; Materials Science and Engineering, Stanford University; Materials Science and Nano-engineering, Mohammed VI Polytechnic University)	TianHe-2 at National Supercomputer Center in Guangzhou

On the Importance of Using Event-Specific Wave Diffusion Rates in Modeling Diffuse Electron Precipitation	4/6/2022	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
New black indium oxide—tandem photothermal CO2-H2 methanol selective catalyst	3/21/2022	Meira, Débora Motta (CLS@APS, Advanced Photon Source, Argonne National Laboratory; Canadian Light Source Inc., 44 Innovation Boulevard, S7N 2V3, Saskatoon, SK, Canada)	National Supercomputer Center in Shenzhen
Incommensurate–commensurate magnetic phase transition in double tungstate Li2Co(WO4)2	3/1/2022	Garlea, V. Ovidiu (Neutron Scattering Division, Oak Ridge National Laboratory), Charlton, T. (Neutron Scattering Division, Oak Ridge National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Kohn anomaly and elastic softening in body-centered cubic molybdenum at high pressure	3/1/2022	Alatas, Ahmet (Argonne National Laboratory), Said, Ayman H. (Argonne National Laboratory)	National Supercomputer Centre in Guangzhou
Effects of EMIC Wave-Driven Proton Precipitation on the Ionosphere	1/29/2022	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Giant anisotropic in-plane thermal conduction induced by Anomalous phonons in pentagonal PdSe2	1/1/2022	Alatas, Ahmet (Advanced Photon Source, Argonne National Laboratory); said, Ayman H. (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Center in Guangzhou
Spin excitations in metallic kagome lattice FeSn and CoSn	11/11/2021	Stone, Matthew B. (Neutron Scattering Division, Oak Ridge National Laboratory), Daemen, Luke L. (Neutron Scattering Division, Oak Ridge National Laboratory), Feng, Erxi (Neutron Scattering Division, Oak Ridge National Laboratory), Cao, Huibo (Neutron Scattering Division, Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou

Significant phase-space-driven thermal transport suppression in BC8 silicon	11/1/2021	Abernathy, Doug (Neutron Scattering Division, Oak Ridge National Laboratory),	National Supercomputer Centre in Guangzhou
Crystal Symmetry Engineering in Epitaxial Perovskite Superlattices	8/26/2021	Jang, Jae Hyuch (Center for Nanophase Materials Sciences Oak Ridge National Laboratory), Gai, Zheng (Center for Nanophase Materials Sciences Oak Ridge National Laboratory), Borisevich, Albina (Center for Nanophase Materials Sciences Oak Ridge National Laboratory)	TianHe-1 at National Supercomputer Center in Tianjin
Simulating the effects of warm O + ions on the growth of electromagnetic ion cyclotron (EMIC) waves	8/15/2021	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Temperature-Induced Structural Changes in the Liquid GaInSn Eutectic Alloy	3/29/2021	Ren, Yang (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Centers in Tianjin
On the Ion Precipitation due to Field Line Curvature (FLC) and EMIC Wave Scattering and Their Subsequent Impact on Ionospheric Electrodynamics	3/12/2021	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Anomalous fast atomic dynamics in bulk metallic glasses	3/1/2021	Dufresne, Eric M. (X-ray Science Division, Advanced Photon Source, Argonne National Laboratory); Ren, Yang (X-ray Science Division, Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Center in Tianjin
Magnetic order and fluctuations in the quasi-two-dimensional planar magnet Sr(Co1−xNix)2As2	12/24/2020	Stone, Matthew B. (Neutron Scattering Division, Oak Ridge National Laboratory), Cao, Huibo (Neutron Scattering Division, Oak Ridge National Laboratory), Abernathy, D. L. (Neutron Scattering Division, Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou

Dynamical Modulation of Solar Flare Electron Acceleration due to Plasmoid-shock Interactions in the Looptop Region	12/16/2020	Guo, Fan (Los Alamos National Laboratory; New Mexico Consortium)	TianHe-2 at National Supercomputer Center in Guangzhou
Temperature-induced structural evolution in liquid Ag-Ga alloys	12/1/2020	Ren, Yang (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Centers in Tianjin
Processing Full-Scale Square Kilometre Array Data on the Summit Supercomputer	11/19/2020	Wang, Ruonan (Computer Science and Mathematics Division, Oak Ridge National Laboratory); Podhorszki, Norbert (Computer Science and Mathematics Division, Oak Ridge National Laboratory); Anantharaj, Valentine (National Center for Computational Sciences, Oak Ridge National Laboratory); Suchyta, Eric (Computer Science and Mathematics Division, Oak Ridge National Laboratory); Klasky, Scott (Computer Science and Mathematics Division, Oak Ridge National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Langevin simulations of the half-filled cubic Holstein model	10/13/2020	Barros, Kipton (Los Alamos National Laboratory, Physics and Chemistry of Materials, Theoretical Division)	National Supercomputer Centers in Tianjin and Guangzhou
Distance-Selected Topochemical Dehydro-Diels–Alder Reaction of 1,4-Diphenylbutadiyne toward Crystalline Graphitic Nanoribbons	9/8/2020	Tulk, Christopher A. (Neutron Scattering Division, Neutron Sciences Directorate, Oak Ridge National Laboratory), Molaison, Jamie J. (Neutron Scattering Division, Neutron Sciences Directorate, Oak Ridge National Laboratory)	TianHe Supercomputer at National Supercomputer Center in Guangzhou
Orbital selective spin waves in detwinned NaFeAs	8/1/2020	Stone, M. B. (Quantum Condensed Matter Division, Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou
The Effects of Field Line Curvature (FLC) Scattering on Ring Current Dynamics and Isotropic Boundary	8/1/2020	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou

Antiferromagnetism of Double Molybdate LiFe(MoO4)2	6/2/2020	Zou, Tao (Neutron Scattering Division, Oak Ridge National Laboratory), Garlea, V. Ovidiu (Neutron Scattering Division, Oak Ridge National Laboratory), Charlton, Timothy (Neutron Scattering Division, Oak Ridge National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Memory-Efficient and Skew-Tolerant MapReduce Over MPI for Supercomputing Systems	5/28/2020	Guo, Yanfei (Mathematics and Computer Science Division, Argonne National Laboratory), Balaji, Pavan (Mathematics and Computer Science Division, Argonne National Laboratory)	National Supercomputer Center in Guangzhou
Anomalously Suppressed Thermal Conduction by Electron-Phonon Coupling in Charge-Density-Wave Tantalum Disulfide	4/23/2020	Alatas, Ahmet (Advanced Photon Source, Argonne National Laboratory); Said, Ayman (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Centre in Guangzhou
Strong local moment antiferromagnetic spin fluctuations in V-doped LiFeAs	2/13/2020	Abernathy, D. L. (Neutron Scattering Division, Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou
The Acceleration and Confinement of Energetic Electrons by a Termination Shock in a Magnetic Trap: An Explanation for Nonthermal Loop-top Sources during Solar Flares	12/20/2019	Guo, Fan (Los Alamos National Laboratory; New Mexico Consortium)	TianHe-1A at National Supercomputer Center in Tianjin and TianHe-2 National Supercomputer Center in Guangzhou
A Universal Seeding Strategy to Synthesize Single Atom Catalysts on 2D Materials for Electrocatalytic Applications	11/14/2019	Yang, Shi-Ze (Materials Science and Technology Division, Oak Ridge National Laboratory),	TianHe-1A at National Supercomputer Center in Tianjin
Power–Law Feature of Structure in Metallic Glasses	11/5/2019	Zuo, X. B. (Advanced Photon Source, Argonne National Laboratory, Xray Sci Div), Ren, Y. (Advanced Photon Source, Argonne National Laboratory, Xray Sci Div)	National Supercomputer Centers in Tianjin and Guangzhou

Structural evolution of low-temperature liquid GaIn eutectic alloy	11/1/2019	Guo, F.M. (Advanced Photon Source, Argonne National Laboratory); Ren, Y. (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Centers in Tianjin
Temperature-Dependent Structural Evolution in Au44Ga56 Liquid Eutectic Alloy	10/4/2019	Ren, Yang (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Centers in Tianjin
The Acceleration of Energetic Particles at Coronal Shocks and Emergence of a Double Power-law Feature in Particle Energy Spectra	9/19/2019	Guo, Fan (Los Alamos National Laboratory; New Mexico Consortium)	TianHe-1A at National Supercomputer Center in Tianjin
The Role of Ru in Improving the Activity of Pd toward Hydrogen Evolution and Oxidation Reactions in Alkaline Solutions	9/13/2019	Xu, Gui-Liang (Chemical Sciences and Engineering Division, Argonne National Laboratory), Zhang, Xiaoyi (X-ray Science Division, Advanced Photon Source, Argonne National Laboratory), Sun, Chengjun (X-ray Science Division, Advanced Photon Source, Argonne National Laboratory), Amine, Khalil (Chemical Sciences and Engineering Division, Argonne National Laboratory; Materials Science and Engineering, Stanford University; IRMC, Imam Abdulrahman Bin Faisal University)	TianHe-2 at National Supercomputer Center in Guangzhou
Frustrated Dipole Order Induces Noncollinear Proper Ferrielectricity in Two Dimensions	8/7/2019	Moreo, Adriana (Materials Science and Technology Division, Oak Ridge National Laboratory), Dagotto, Elbio (Materials Science and Technology Division, Oak Ridge National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Pressure effect on Kohn anomaly and electronic topological transition in single-crystal tantalum	8/1/2019	Alatas, Ahmet (Advanced Photo Source, Argonne National Laboratory), Said, Ayman H. (Advanced Photo Source, Argonne National Laboratory)	National Supercomputer Centre in Guangzhou

Low-temperature anharmonicity and the thermal conductivity of cesium iodide	5/1/2019	Chi, Songxue (Neutron Scattering Division, Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou
Coexistence of Ferromagnetic and Stripe Antiferromagnetic Spin Fluctuations in SrCo2As2	3/21/2019	Abernathy, D. L. (Neutron Scattering Division, Oak Ridge National Laboratory), Niedziela, J. L. (Neutron Scattering Division, Oak Ridge National Laboratory), Pajerowski, Daniel M. (Neutron Scattering Division, Oak Ridge National Laboratory), Matsuda, Masaaki (Neutron Scattering Division, Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou
Pressure-Induced Diels–Alder Reactions in C6H6-C6F6 Cocrystal towards Graphane Structure	1/3/2019	Chen, Jihua (Center for Nanophase Materials Sciences, Oak Ridge National Laboratory), Hensley, Dale K. (Center for Nanophase Materials Sciences, Oak Ridge National Laboratory), Ren, Yang (Advanced Photon Source, Argonne National Laboratory)	TianHe Supercomputer at National Supercomputer Center in Guangzhou
On the Power of Combiner Optimizations in MapReduce Over MPI Workflows	12/11/2018	Guo, Yanfei (Argonne National Laboratory), Balaji, Pavan (Argonne National Laboratory)	National Supercomputer Center in Guangzhou
Charge transfer drives anomalous phase transition in ceria	11/29/2018	Ren, Yang (X-Ray Science Division, Argonne National Laboratory); Neuefeind, Joerg C. (Chemical and Engineering Materials Division, Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou
Coexistence of Slow-mode and Alfvén-mode Waves and Structures in 3D Compressive MHD Turbulence	10/8/2018	Li, Shengtai (Theoretical Division, MS B284, Los Alamos National Laboratory)	TianHe-1A at National Supercomputer Center in Tianjin
c-axis pressure-induced antiferromagnetic order in optimally P-doped BaFe2(As0.70P0.30)2 superconductor	9/18/2018	Cao, Huibo (Neutron Scattering Division, Oak Ridge National Laboratory); Tian, Wei (Neutron Scattering Division, Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou

Tailored Synthesis of the Narrowest Zigzag Graphene Nanoribbon Structure by Compressing the Lithium Acetylide under High Temperature	8/13/2018	Meng, Yue (HPCAT, X-Ray Science Division, Argonne National Laboratory)	National Supercomputer Center in Guangzhou
Magnetic properties of L10 FePt thin film influenced by recoverable strains stemmed from the polarization of Pb(Mg1/3Nb2/3)O3–PbTiO3 substrate* *Project supported by the National Natural Science Foundation of China (Grant Nos. 11474167, 51701091, and 11504020) and the Start-up Funds of NUIST, China (Grant Nos. 2243141601035 and 2243141601018).	7/16/2018	Liang, Linyun (Mathematics and Computer Science Division, Argonne National Laboratory)	National Supercomputer Center in Guangzhou
Self-Consistent Modeling of Electron Precipitation and Responses in the Ionosphere: Application to Low-Altitude Energization During Substorms	7/12/2018	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory),	TianHe-2 at National Supercomputer Center in Guangzhou
Disappearance of Anisotropic Intermittency in Large-amplitude MHD Turbulence and Its Comparison with Small-amplitude MHD Turbulence	3/8/2018	Li, Shengtai (Theoretical Division, MS B284, Los Alamos National Laboratory)	TianHe-1A at National Supercomputer Center in Tianjin
Sequential structural and antiferromagnetic transitions in BaFe2Se3 under pressure	1/1/2018	Dagotto, Elbio (Materials Science and Technology Division, Oak Ridge National Laboratory; Department of Physics and Astronomy, University of Tennessee)	TianHe-2 at National Supercomputer Center in Guangzhou
Formation and Properties of Tangential Discontinuities in Three-dimensional Compressive MHD Turbulence	12/19/2017	Li, Shengtai (Theoretical Division, MS B284, Los Alamos National Laboratory)	TianHe-1A at National Supercomputer Center in Tianjin
Bloomfish: A Highly Scalable Distributed K-Mer Counting Framework	12/1/2017	Guo, Yanfei (Argonne National Laboratory), Balaji, Pavan (Argonne National Laboratory)	National Supercomputer Center in Guangzhou

Influence of Intermittency on the Quasi-perpendicular Scaling in Three-dimensional Magnetohydrodynamic Turbulence	8/30/2017	Li, Shengtai (Theoretical Division, MS B284, Los Alamos National Laboratory)	TianHe-1A at National Supercomputer Center in Tianjin
PIC simulations of wave-particle interactions with an initial electron velocity distribution from a kinetic ring current model	7/15/2017	Delzanno, Gian Luca (Los Alamos National Laboratory), Jordanova, Vania (Los Alamos National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
DALiuGE: A graph execution framework for harnessing the astronomical data deluge	7/1/2017	Wang, R. (Oak Ridge National Laboratory)	National Supercomputer Center in Guangzhou
Liquid-to-liquid crossover in the GaIn eutectic alloy	6/1/2017	Ren, Y. (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Centers in Tianjin and Guangzhou
Climate mitigation from vegetation biophysical feedbacks during the past three decades	5/22/2017	Mao, Jiafu (Environmental Sciences Division and Climate Change Science Institute, Oak Ridge National Laboratory), Shi, Xiaoying (Environmental Sciences Division and Climate Change Science Institute, Oak Ridge National Laboratory)	National Supercomputer Center in Tianji
Effects of electric field methods on modeling the midlatitude ionospheric electrodynamics and inner magnetosphere dynamics	5/22/2017	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory),	TianHe-2 at National Supercomputer Center in Guangzhou
Cycloidal magnetism driven ferroelectricity in double tungstate LiFe(WO4)2	5/1/2017	Garlea, V. Ovidiu (Quantum Condensed Matter Division, Oak Ridge National Laboratory), Zou, Tao (Quantum Condensed Matter Division, Oak Ridge National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Temperature-dependent structure evolution in liquid gallium	4/1/2017	Ren, Y. (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Centers in Tianjin and Guangzhou
Pressure-driven phase transition from antiferromagnetic semiconductor to	3/1/2017	Dagotto, Elbio (Materials Science and Technology Division, Oak Ridge National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou

nonmagnetic metal in the two-leg ladders AFe2X3 (A=Ba,K; X=S,Se)			
Structural evolution and dynamical properties of Al2Ag and Al2Cu liquids studied by experiments and ab initio molecular dynamics simulations	3/1/2017	Guo, F.M. (Advanced Photon Source, Argonne National Laboratory); Ren, Y. (Advanced Photon Source, Argonne National Laboratory)	National Supercomputer Centers in Tianjin and Guangzhou
Multiscale Pressure-Balanced Structures in Three-dimensional Magnetohydrodynamic Turbulence	2/8/2017	Li, Shengtai (Theoretical Division, MS B284, Los Alamos National Laboratory)	TianHe-1A at National Supercomputer Center in Tianjin
Anomalously low electronic thermal conductivity in metallic vanadium dioxide	1/27/2017	Hong, Jiawang (Materials Science and Technology Division, Oak Ridge National Laboratory), Delaire, Olivier (Materials Science and Technology Division, Oak Ridge National Laboratory; Department of Mechanical Engineering and Materials Science, Duke University)	National Supercomputer Center in Guangzhou
Interplay of Cation Ordering and Ferroelectricity in Perovskite Tin Iodides: Designing a Polar Halide Perovskite for Photovoltaic Applications	9/28/2016	Rondinelli, James M. (Materials Science Division, Argonne National Laboratory)	National Supercomputer Center in Tianjin
A new ionospheric electron precipitation module coupled with RAM-SCB within the geospace general circulation model	9/17/2016	Jordanova, Vania K. (Space Science and Application, Los Alamos National Laboratory), Jeffery, Christopher A. (Space Science and Application, Los Alamos National Laboratory)	TianHe-2 at National Supercomputer Center in Guangzhou
Fast Diffusion of Native Defects and Impurities in Perovskite Solar Cell Material CH3NH3PbI3	6/9/2016	Ming, Wenmei (Materials Science and Technology Division, Oak Ridge National Laboratory), Shi, Hongliang (Materials Science and Technology Division, Oak Ridge National Laboratory; Key Laboratory of Micro-Nano Measurement-Manipulation and Physics (Ministry of Education), School of Physics and Nuclear Energy Engineering, Beihang	TianHe-1A at National Supercomputer Center in Tianjin

		University), Du, Mao-Hua (Materials Science and Technology Division, Oak Ridge National Laboratory)	